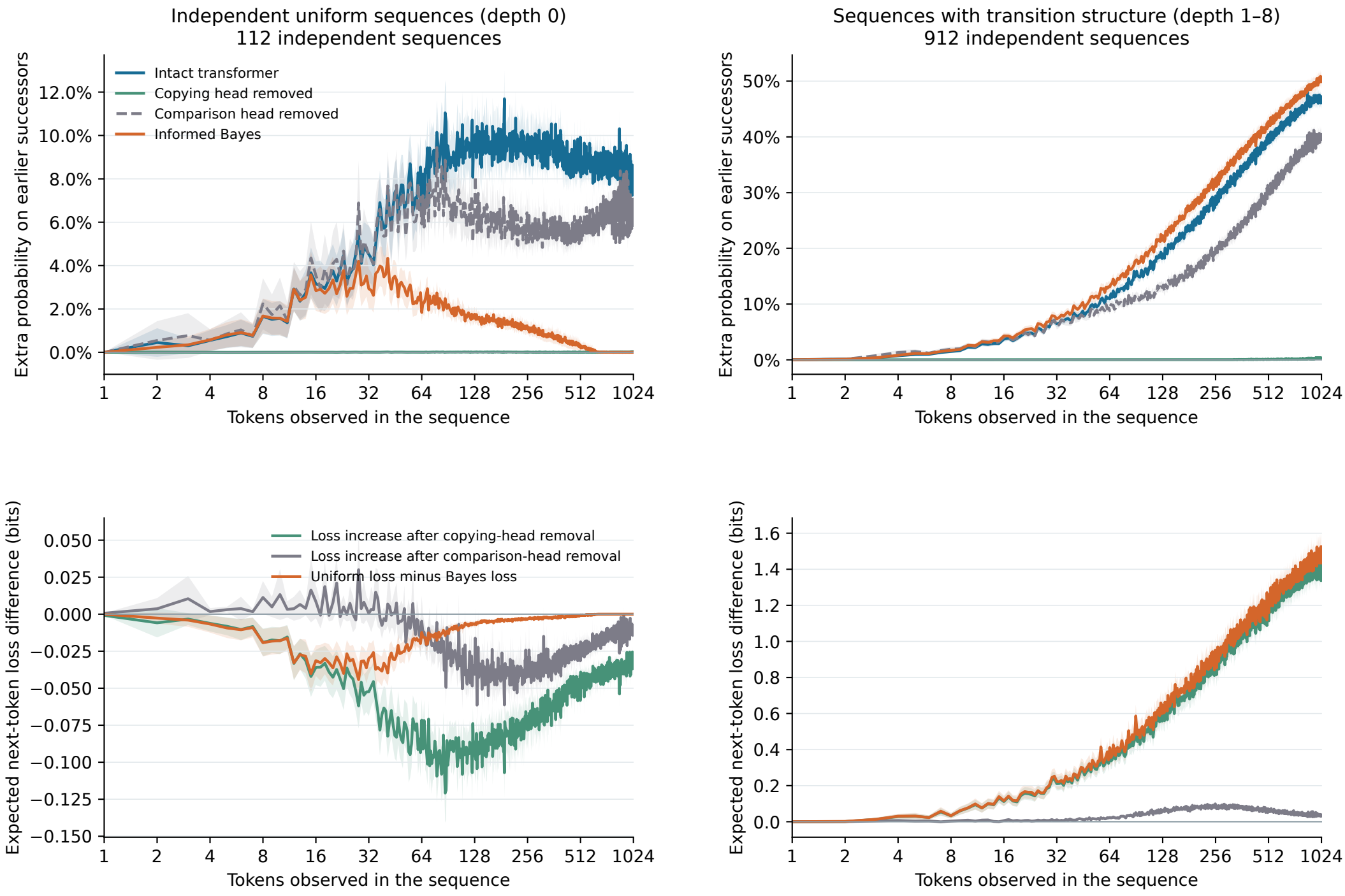

Source controls and predictive benefit
Independent uniform sequences (depth 0)
112 independent sequences
Sequences with transition structure (depth 1–8)
912 independent sequences
Intact transformer
Copying head removed
Comparison head removed
Informed Bayes
Extra probability on earlier successors
Tokens observed in the sequence
Loss increase after copying-head removal
Loss increase after comparison-head removal
Uniform loss minus Bayes loss
Expected next-token loss difference (bits)
Head removals are compared with the intact transformer; Bayes is compared with the known uniform distribution.
All signed values are retained. Shading: pointwise 95% confidence intervals across independent sequences; curves are unsmoothed.